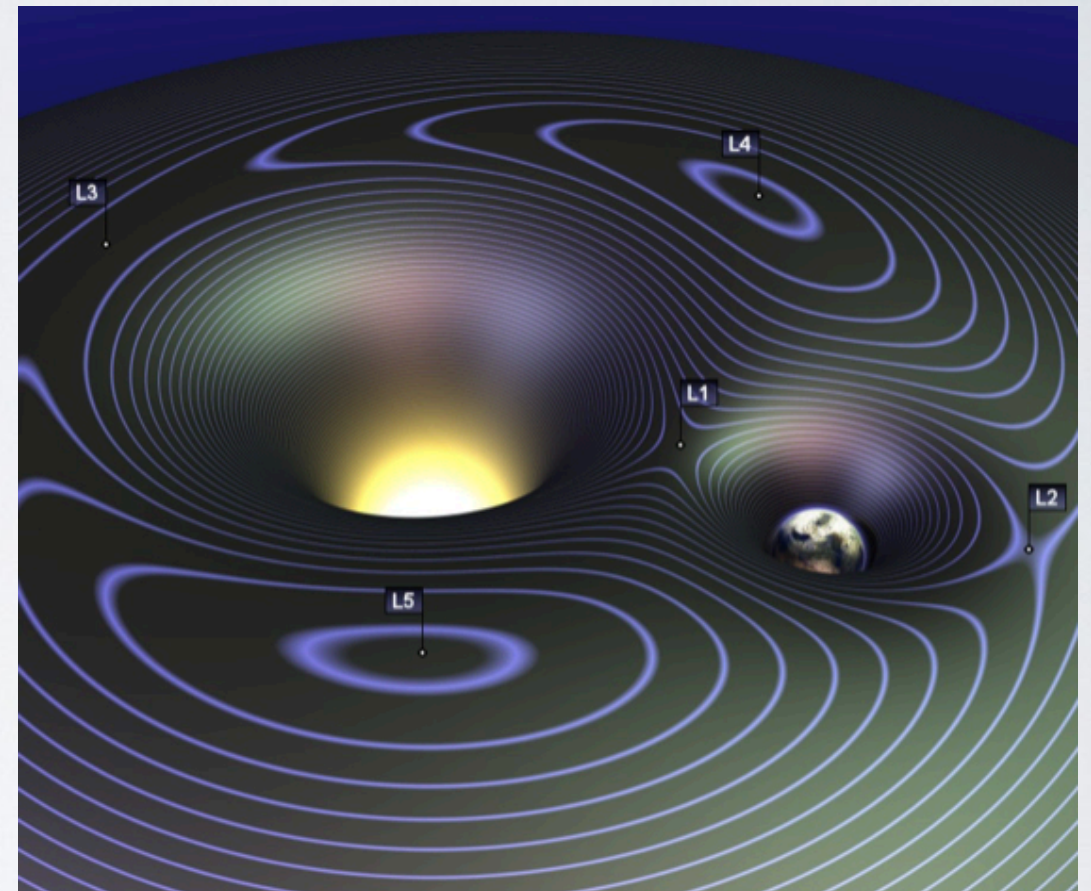
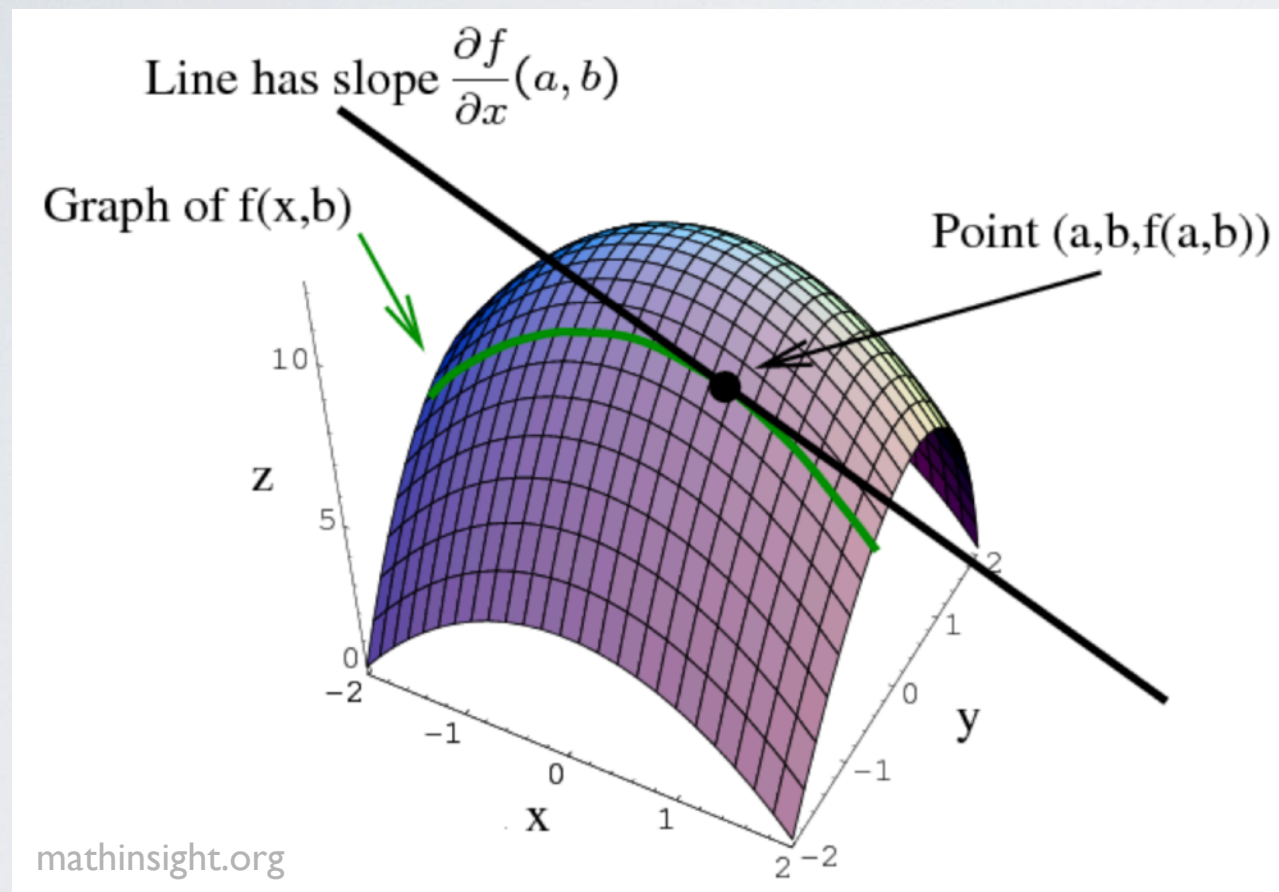


Week 5 - #2

Partial Differentiation



Today: Ch 4

Next Class: Ch 5

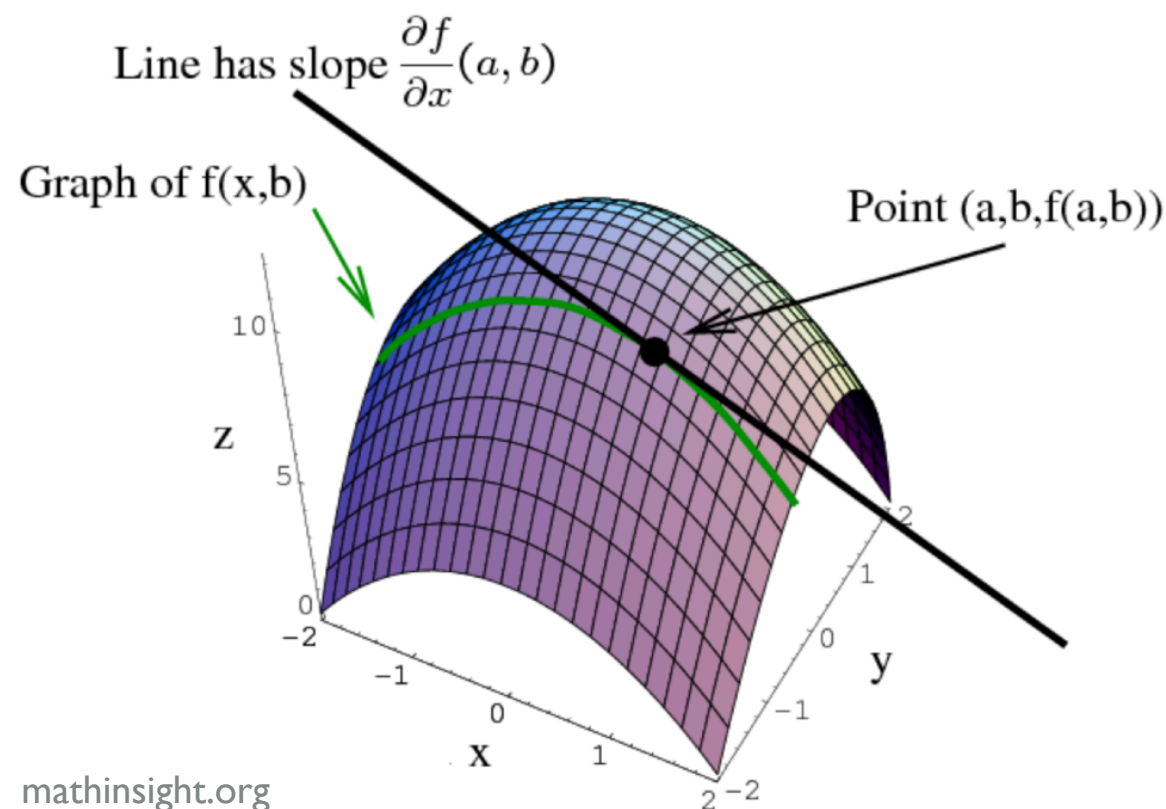
Ji-hoon Kim (Seoul National University)

Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #7

— [open book and open note, “and” cellphone or laptop, drop it off as you leave the class] —

Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Boas, Chapter 4, Section 4, Problem 1
2. What is the difference between the two formulae, Eq.(3.6) and (5.3) of Boas, Chapter 4?
3. List any research area you can think of where tensors are used — e.g., in physics textbooks, or in describing daily phenomena of any kind.



Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 6, Boas, 3rd ed.

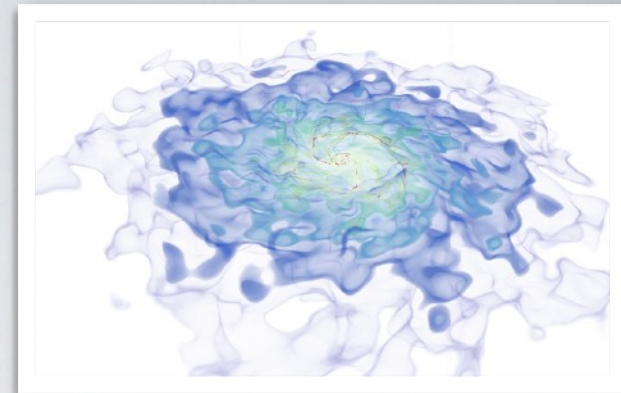
The problems I suggest you to take a deeper look into include, but are not limited to, the following.
The class homework assignments will mainly be from this list.

- Section 03: Problems 12, 13, 14, 16, 17
- Section 04: Problems 6, 9
- Section 06: Problems 8, 13, 14, 16
- Section 07: Problems 6, 7, 15, 16, 17
- Section 08: Problems 6, 14, 18, 19, 20
- Section 09: Problems 3, 4, 7, 8
- Section 10: Problems 6, 10, 12, 15, 16
- Section 11: Problems 8, 10, 12, 16, 17(b)(d)(e), 18
- Section 12: Problems 3, 16, 19

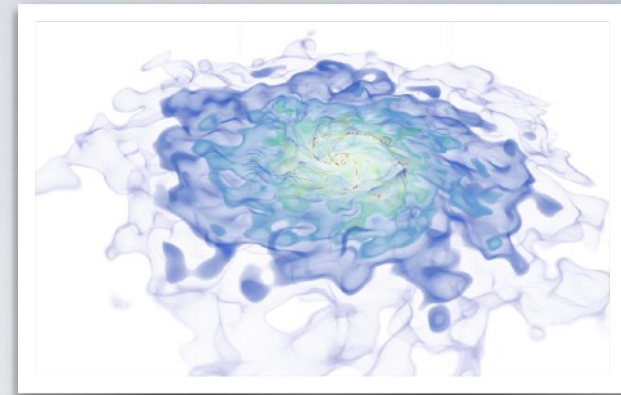
Chapter 10, Sections 5 & 7-9

- Section 05: Problems 5, 7, 10
- Section 08: Problems 6
- Section 09: Problems 2, 5

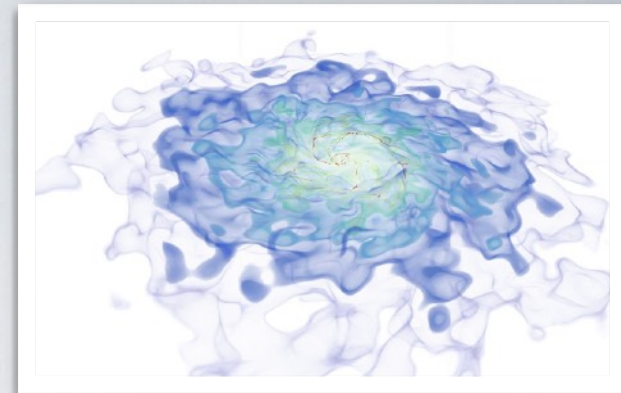
39 problems / 173 total
(Ch.10) 13 problems / 138 total



No Class Next Monday (Oct. 5)



**Recorded Lecture Next Wednesday
(Oct. 7)**



Inertia Tensors

Matrix Diagonalization (Ch 10.4, Ex 4)



eigenvector {{4,2,0}, {2,3,-1}, {0,-1,5}}

NATURAL LANGUAGE

MATH INPUT

EXTENDED KEYBOARD

EXAMPLES

UPLOAD

RANDOM

Input

eigenvectors

$$\begin{pmatrix} 4 & 2 & 0 \\ 2 & 3 & -1 \\ 0 & -1 & 5 \end{pmatrix}$$

Results

Approximate forms

☒ Step-by-step solution

$$v_1 = (-1, -1, 1)$$

$$v_2 = (-1 + \sqrt{3}, 2 - \sqrt{3}, 1)$$

$$v_3 = (-1 - \sqrt{3}, 2 + \sqrt{3}, 1)$$

Corresponding eigenvalues

Enlarge

Data

Customize

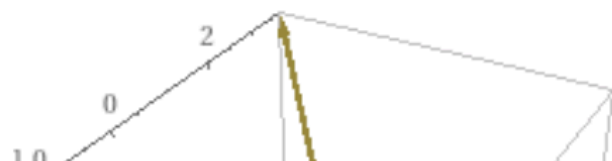
Plain Text

$$\lambda_1 = 6$$

$$\lambda_2 = 3 + \sqrt{3}$$

$$\lambda_3 = 3 - \sqrt{3}$$

Plot of eigenvectors



eigenvector $\{\{4,2,0\}, \{2,3,-1\}, \{0,-1,5\}\}$



NATURAL LANGUAGE

MATH INPUT

EXTENDED KEYBOARD

EXAMPLES

UPLOAD

RANDOM

Input

eigenvectors

$$\begin{pmatrix} 4 & 2 & 0 \\ 2 & 3 & -1 \\ 0 & -1 & 5 \end{pmatrix}$$

Results

Approximate forms

☒ Step-by-step solution

$$v_1 = (-1, -1, 1)$$

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Corresponding eigenvalues

Enlarge

Data

Customize

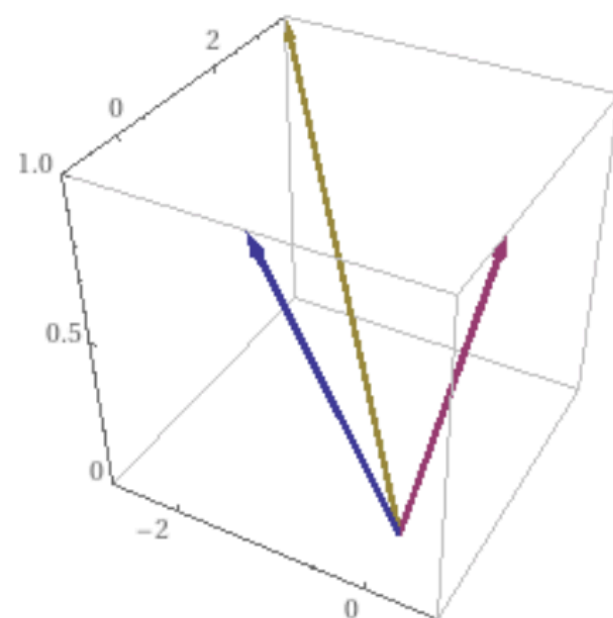
Plain Text

$$\lambda_1 = 6$$

$$\lambda_2 = 3 + \sqrt{3}$$

$$\lambda_3 = 3 - \sqrt{3}$$

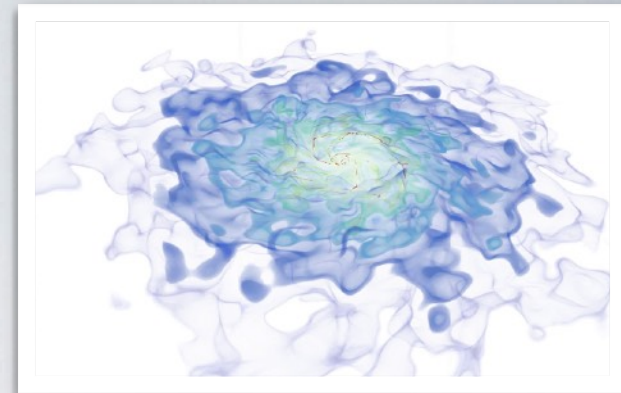
Plot of eigenvectors



— $(-1, -1, 1)$

— $(-1 + \sqrt{3}, 2 - \sqrt{3}, 1)$

— $(-1 - \sqrt{3}, 2 + \sqrt{3}, 1)$



Pitfalls of Blind Computation (Again)

Pitfalls of Blind Computation

- Inaccurate estimation on Google:

The screenshot shows a Google search interface. The search bar contains the expression $\frac{1}{\sqrt{0.25-1e-20}} - \frac{1}{\sqrt{0.25}}$. Below the search bar, navigation links for All, Images, News, Shopping, Maps, and More are visible, along with Settings and Tools. The search results indicate "About 14,900,000 results (0.63 seconds)". Below the results, a calculator interface is displayed. The calculator's display shows the expression $(1 / \sqrt{0.25 - 1e-20}) - (1 / \sqrt{0.25}) =$ followed by the result 0. The calculator interface includes buttons for Rad, Deg, x!, (,), %, AC, Inv, sin, ln, 7, 8, 9, ÷, π, cos, log, 4, 5, 6, ×, e, tan, √, 1, 2, 3, −, Ans, EXP, x^y, 0, ., =, and +.

Pitfalls of Blind Computation

- Inaccurate estimation on Python:

```
bash-3.2$ python
Python 2.7.9 (default, Sep  9 2015, 22:07:15)
[GCC 4.2.1 Compatible Apple LLVM 6.1.0 (clang-602.0.53)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> import math
>>> 1/math.sqrt(0.25-1e-20) - 1/math.sqrt(0.25)
0.0
>>> 1/math.sqrt(0.25-1e-10) - 1/math.sqrt(0.25)
4.000000330961484e-10
>>>
```

Pitfalls of Blind Computation

- Accurate estimation on WolframAlpha:

 **WolframAlpha**[®] computational intelligence[™]

[Extended Keyboard](#) [Upload](#) [Examples](#) [Random](#)

Input interpretation:

$$\frac{1}{\sqrt{0.25 - 1 \times 10^{-20}}} - \frac{1}{\sqrt{0.25}}$$

Result:

$4.00000... \times 10^{-20}$

[More digits](#)

Number line:



A horizontal number line with tick marks at -0.5, 0.0, and 0.5. A blue dot is placed exactly at the 0.0 mark.

Number name:

zero point zero zero zero zero zero zero zero zero
zero zero zero zero zero zero zero zero zero zero zero four

[Download Page](#)

POWERED BY THE WOLFRAM LANGUAGE